

Практикум D2:archive

Every past-paper question in which the answer comes out of the structure of a probability space, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in Practicum D2, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

О практикуме

Первый практикум секции D и первый по статистике вообще. Тема `statistics.probability` — 77 блоков и 237 баллов; D2 берёт 50 блоков и 156 баллов, то есть всё, где ответ получают из устройства вероятностного пространства. Доля 0.658 — это те самые 156 из 237, а не подобранное число.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d2-probability

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D2 archive: probability

Every past-paper question in which the answer comes out of the structure of a probability space, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in *Practicum D2*, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

What is inside. The part of `statistics.probability` that is about events, conditioning, independence and trees, sessions May 2021 — November 2025: **50 questions, 156 marks**, in ten sections, one section per technique.

The whole topic is 77 blocks and 237 marks. The 81 marks left over are mostly not this topic at all: 36 are the binomial distribution (D3), 11 a probability generating function (D4), 7 the normal distribution (D5), 11 a density (D6), and 4 are the November 2023 zonal duplicate

counted twice. The last 12 are descriptive statistics — a box plot, a mean, a sampling method — and they belong to no practicum in the map at all.

How to work. Read the question, answer in the cell below it, run the cell. None of the checks here knows the answer. A question about probability fixes a **space**, and the space fixes the answer: `verify_event` takes the conditions the question states — $P(A) = 0.65$, $P(A \mid B) = \frac{1}{4}$, *A and B are independent* — solves them as equations on the weights of the four cells of the Venn diagram, and computes what was asked from the definition. `verify_probability` is handed the space written out — a tree, a sequence of draws without replacement, 216 equally likely triples — and adds the weights of the outcomes you want; before anything else it checks that the weights sum to one, which is where a mis-copied tree gets caught.

When your answer is wrong they do more than say so. The slips in this topic have names, and each one is rebuilt from the same space: the conditional taken the wrong way round, the union added without subtracting the intersection, independence assumed where the question never promised it, the intersection handed in before it was divided by anything.

Nothing is stored. Not one of the fifty answers is written down in this notebook, as a number or as a hash. Every one of them is worked out from the question every time you run the cell.

One question has no cell — *explain why $Y \leq 5$* — because there is nothing to hand over. Read it, do it on paper, read the solution.

Leave a cell blank and it prints  and moves on, so you can run the whole notebook top to bottom on the first open and nothing breaks.

The solutions are at the end, numbered the same way. Open one after you have worked the question, not before.

The ten sections.

§	technique	questions	marks
1	The algebra of events	8	17
2	Conditional probability	7	22
3	Independence as an equation	5	17
4	Trees	4	10
5	Repeated trials and the first success	7	17
6	The law of total probability	3	11
7	Bayes: back up the tree	5	20
8	Without replacement	3	12
9	Counting an equally likely space	5	18
10	A letter inside the probability	3	12

Sections 1–3 live on a Venn diagram: two events, four cells, and every question is arithmetic once the cells are filled.

Sections 4, 5, 8 and 9 build the space first — that is where the real work of this topic is. Sections 6 and 7 are one machine run in two directions: down the tree and back up it. Section 10 puts a letter in the space and asks you to find it, and it is the only place in the topic where an answer is thrown away for being bigger than one.

62% of the marks carry a calculator, and here that is nearly honest.

Genuine GDC work comes to about 30 marks out of 156 — the normal and binomial probabilities that sit on the branches of the trees in §6 and §7. The other 126 are done on paper.

```

import sys
sys.path.append('..')          # from
practicum/statistics to practicum/kit.py
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Rational, Eq, events, P(...)

language('en')                # this
notebook is in English, and so are the checks

a_, b_, k_, p_, r_, y_, z_ = symbols('a b k p
r y z')

print('ready; sympy', sp.__version__)
print('a probability: ', Rational(9, 14))
print('two numbers:   ', [Rational(1, 6),
Rational(1, 6)])
print('an event:      ', events('A B'))

```

1. The algebra of events

$$\begin{aligned}
 P(A \cup B) &= P(A) \\
 &+ P(B) \\
 &- P(A \cap B), \quad P(A' \cap B') \\
 &= 1 - P(A \cup B)
 \end{aligned}$$

Two events cut the sample space into four cells: $A \cap B$, $A \cap B'$, $A' \cap B$, $A' \cap B'$. Their probabilities add to one, and every question in this section is arithmetic

once you know all four. Draw the diagram before you write anything — it is faster than the formula and it does not let you forget the fourth cell.

1.1 — November 2023 TZ1 Paper 1 Q2(a), 2 marks

Events A and B are such that $P(A) = 0.65$, $P(B) = 0.75$ and $P(A \cap B) = 0.6$.

Find $P(A \cup B)$.

```
A, B = events('A B')
here = [(P(A), 0.65), (P(B), 0.75), (P(A &
B), 0.6)]

q1_1 = ...

verify_event('1.1', q1_1, here, P(A | B))
```

1.2 — November 2023 TZ1 Paper 1 Q2(b), 2 marks

Hence, or otherwise, find $P(A' \cap B')$.

```
q1_2 = ...

verify_event('1.2', q1_2, here, P(~A & ~B))
```

1.3 — May 2021 TZ2 Paper 2 Q3(a), 2 marks

At a school, 70% of the students play a sport and 20% of the students are involved in theatre. 18% of the students do neither activity.

A student is selected at random. Find the probability that the student plays a sport and is involved in theatre.

```
S, T = events('S T')
school = [(P(S), Rational(7, 10)), (P(T),
Rational(2, 10)),
          (P(~S & ~T), Rational(18, 100))]

q1_3 = ...

verify_event('1.3', q1_3, school, P(S & T))
```

1.4 — May 2021 TZ2 Paper 2 Q3(b), 2 marks

Find the probability that the student is involved in theatre, but does not play a sport.

```
q1_4 = ...

verify_event('1.4', q1_4, school, P(T & ~S))
```

1.5 — May 2025 TZ2 Paper 1 Q4(a), 3 marks

Events A and B are such that $P(A \cup B) = \frac{5}{8}$ and $P(A \cap B') = \frac{7}{24}$.

Find $P(B)$.

```
A, B = events('A B')
pair = [(P(A | B), Rational(5, 8)), (P(A &
~B), Rational(7, 24))]

q1_5 = ...

verify_event('1.5', q1_5, pair, P(B))
```

1.6 — November 2022 Paper 1 Q6(a), 1 mark

Events A and B are such that $P(A) = 0.3$ and $P(B) = 0.8$.

Determine the value of $P(A \cap B)$ in the case where the events A and B are independent.

```
A, B = events('A B')
loose = [(P(A), Rational(3, 10)), (P(B),
Rational(8, 10))]

q1_6 = ...

verify_event('1.6', q1_6, loose + [Eq(P(A &
B), P(A)*P(B))], P(A & B))
```

1.7 — November 2022 Paper 1 Q6(b), 3 marks

Determine the minimum possible value of $P(A \cap B)$.

Two probabilities do not fix a space. The question is asking how small the overlap can be made while everything stays a probability, and the check answers it the same way: it looks for the smallest value $P(A \cap B)$ takes over every space the two conditions allow.

```
q1_7 = ...

verify_event('1.7', q1_7, loose, P(A & B),
extreme='min')
```

1.8 — November 2022 Paper 1 Q6(c), 2 marks

Determine the maximum possible value of $P(A \cap B)$, justifying your answer.

The number is checkable; the justification is not, and the markscheme is strict about it — «Do not award R0A1». The solution says what has to be written.

```
q1_8 = ...
```

```
verify_event('1.8', q1_8, loose, P(A & B),  
extreme='max')
```

2. Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The bar cuts the space down to B and asks the question again inside it. Two things go wrong here and they go wrong constantly: the two events swap places, and the division never happens. Both are named by the check, so when you get one of them you will be told which.

2.1 — November 2025 TZ3 Paper 1

Q2(a), 2 marks

Events A and B are such that $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A | B) = \frac{1}{4}$.

Find the probability that both events A and B occur.

```
A, B = events('A B')
given = [(P(A), Rational(1, 2)), (P(B),
Rational(1, 3)),
(P(A, given=B), Rational(1, 4))]

q2_1 = ...

verify_event('2.1', q2_1, given, P(A & B))
```

2.2 — November 2025 TZ3 Paper 1

Q2(b), 4 marks

Find $P(A | B')$.

```
q2_2 = ...

verify_event('2.2', q2_2, given, P(A,
given=~B))
```

2.3 — May 2025 TZ1 Paper 1 Q4(a), 3 marks

Consider events A and B such that $P(A') = P(A \cup B) = \frac{3}{4}$ and $P(B | A) = \frac{2}{3}$.

Find $P(A \cap B)$.

```
A, B = events('A B')
may25 = [(P(~A), Rational(3, 4)), (P(A | B),
Rational(3, 4)),
          (P(B, given=A), Rational(2, 3))]

q2_3 = ...

verify_event('2.3', q2_3, may25, P(A & B))
```

2.4 — May 2023 TZ2 Paper 1 Q3, 5 marks

Events A and B are such that $P(A) = 0.4$, $P(A | B) = 0.25$ and $P(A \cup B) = 0.55$.

Find $P(B)$.

```
A, B = events('A B')
two = [(P(A), Rational(4, 10)), (P(A,
given=B), Rational(25, 100)),
        (P(A | B), Rational(55, 100))]

q2_4 = ...

verify_event('2.4', q2_4, two, P(B))
```

2.5 — May 2022 TZ1 Paper 2 Q5(b), 2 marks

Let A and B be two independent events such that $P(A \cap B') = 0.16$ and $P(A' \cap B) = 0.36$.

Find $P(A' \mid B')$.

```
A, B = events('A B')
indep = [Eq(P(A & B), P(A)*P(B)),
         (P(A & ~B), Rational(16, 100)),
         (P(~A & B), Rational(36, 100))]

q2_5 = ...

verify_event('2.5', q2_5, indep, P(~A,
given=~B))
```

2.6 — May 2025 TZ2 Paper 1 Q4(b), 4 marks

Events A and B are such that $P(A \cup B) = \frac{5}{8}$ and $P(A \cap B') = \frac{7}{24}$.

Given that events A and B are independent, find $P(A' \mid B)$.

```

A, B = events('A B')
same = [(P(A | B), Rational(5, 8)), (P(A &
~B), Rational(7, 24)),
        Eq(P(A & B), P(A)*P(B))]

q2_6 = ...

verify_event('2.6', q2_6, same, P(~A,
given=B))

```

2.7 — May 2021 TZ2 Paper 2 Q3(c), 2 marks

At the school 48% of the students are girls, and 25% of the girls are involved in theatre.

A student is selected at random. Let G be the event "the student is a girl" and let T be the event "the student is involved in theatre".

Find $P(G \cap T)$.

```

G, T = events('G T')
girls = [(P(G), Rational(48, 100)), (P(T),
Rational(2, 10)),
        (P(T, given=G), Rational(25, 100))]

q2_7 = ...

verify_event('2.7', q2_7, girls, P(G & T))

```

3. Independence as an equation

$$\begin{aligned} A \text{ and } B \text{ are independent} &\iff P(A \cap B) \\ &= P(A)P(B) \end{aligned}$$

That is a definition, not a property, and it works in both directions. When the question asks whether two events are independent, you compute both sides and compare. When the question *tells* you they are independent, you have gained an equation you did not have before — usually the one that makes the question solvable at all.

The first three questions here ask for the verdict. You hand over the two numbers the comparison rests on, in the order $P(A)P(B)$ then $P(A \cap B)$ — which is exactly what the markscheme wants to see, and it will not give the mark for a conclusion with no numbers under it.

3.1 — *May 2025 TZ1 Paper 1 Q4(b), 3 marks*

Consider events A and B such that $P(A') = P(A \cup B) = \frac{3}{4}$ and $P(B \mid A) = \frac{2}{3}$.

Show that events A and B are independent.

```
A, B = events('A B')
may25 = [(P(~A), Rational(3, 4)), (P(A | B),
Rational(3, 4)),
(P(B, given=A), Rational(2, 3))]

q3_1 = ... # [P(A)*P(B), P(A and B)]

verify_independence('3.1', q3_1, may25, A, B)
```

3.2 — May 2021 TZ2 Paper 2 Q3(d), 2 marks

At the school 20% of the students are involved in theatre, 48% are girls, and 25% of the girls are involved in theatre.

Determine if the events G and T are independent. Justify your answer.

```
G, T = events('G T')
girls = [(P(G), Rational(48, 100)), (P(T),
Rational(2, 10)),
(P(T, given=G), Rational(25, 100))]

q3_2 = ... # [P(G)*P(T), P(G and T)]

verify_independence('3.2', q3_2, girls, G, T)
```

3.3 — November 2021 Paper 1 Q4(b), 2 marks

Box 1 contains 5 red balls and 2 white balls. Box 2 contains 4 red balls and 3 white balls. A box is chosen at random and a ball is drawn.

Let A be the event that "box 1 is chosen" and let R be the event that "a red ball is drawn".

Determine whether events A and R are independent.

```
A, Rd = events('A R')
boxes = [(P(A), Rational(1, 2)),
         (P(Rd, given=A), Rational(5, 7)),
         (P(Rd, given=~A), Rational(4, 7))]

q3_3 = ... # [P(A)*P(R), P(A and R)]

verify_independence('3.3', q3_3, boxes, A,
Rd)
```

3.4 — May 2022 TZ1 Paper 2 Q5(a), 4 marks

Let A and B be two independent events such that $P(A \cap B') = 0.16$ and $P(A' \cap B) = 0.36$.

Given that $P(A \cap B) = x$, find the value of x .

Here independence is the third equation. Without it the two numbers given leave the space undetermined; with it they pin it exactly — and the equation it adds is quadratic, which is why this part is worth four marks and the next one two.

```
A, B = events('A B')
indep = [Eq(P(A & B), P(A)*P(B)),
         (P(A & ~B), Rational(16, 100)),
         (P(~A & B), Rational(36, 100))]

q3_4 = ...

verify_event('3.4', q3_4, indep, P(A & B))
```

3.5 — May 2022 TZ2 Paper 2 Q3, 6 marks

Events A and B are independent and $P(A) = 3P(B)$.

Given that $P(A \cup B) = 0.68$, find $P(B)$.

The quadratic has two roots and the markscheme takes only one: «Award A1 if both answers are given as final answers for $P(B)$ ». The other root is $\frac{17}{15}$,

and what is wrong with it is not that it is extraneous — it is that it is not a probability.

```
A, B = events('A B')
triple = [Eq(P(A & B), P(A)*P(B)),
          Eq(sympify(P(A)), 3*sympify(P(B))),
          (P(A | B), Rational(68, 100))]

q3_5 = ...

verify_event('3.5', q3_5, triple, P(B))
```

4. Trees

**Along a branch you multiply;
between branches you add.**

A tree is the right picture whenever the experiment has stages. The first level carries the probabilities of the first stage and adds to one; every node of the second level carries *conditional* probabilities and adds to one on its own. From here on the space is written out as a dictionary of outcomes and weights, and the check adds up the ones you want — after first making sure your weights

come to one, which is where a mis-copied tree is caught.

4.1 — May 2023 TZ1 Paper 2 Q7(a), 2 marks

A new test has been developed to identify whether a particular gene, G , is present in a population of parrots. The test returns a correct positive result 95% of the time for parrots with the gene, and a false positive result 2% of the time for parrots without the gene. The proportion of the population with the gene is p .

Complete the tree diagram: the branch from the root to G' , and the two branches out of G .

Three boxes, and the check asks for the whole tree at once: write the weight of each of the four paths and it will tell you whether they come to one and whether the branch out of G is right. The markscheme is strict about the first box — «award A0 for G' branch labelled q unless explicitly defined as $1 - p$ ».

```

q4_1 = ...          # the four path weights,
in order (G,+), (G,-), (G',+), (G',-)

verify_constants('4.1', q4_1,
symbols('u1:5'), [
    ('the four paths together are certain',
Eq(sum(symbols('u1:5')), 1)),
    ('a proportion p of the parrots carry the
gene',
Eq(symbols('u1') + symbols('u2'), p_)),
    ('the test is right 95% of the time on
parrots that carry it',
Eq(symbols('u1'), Rational(95,
100)*p_)),
    ('and gives a false positive 2% of the
time on parrots that do not',
Eq(symbols('u3'), Rational(2, 100)*(1 -
p_))),
])

```

4.2 — November 2021 Paper 1 Q4(a), 3 marks

Box 1 contains 5 red balls and 2 white balls. Box 2 contains 4 red balls and 3 white balls.

A box is chosen at random and a ball is drawn. Find the probability that the ball is red.

```
boxes = {('1', 'red'): Rational(1,
2)*Rational(5, 7),
        ('1', 'white'): Rational(1,
2)*Rational(2, 7),
        ('2', 'red'): Rational(1,
2)*Rational(4, 7),
        ('2', 'white'): Rational(1,
2)*Rational(3, 7)}

q4_2 = ...

verify_probability('4.2', q4_2, boxes, lambda
o: o[1] == 'red')
```

4.3 — November 2022 Paper 2 Q10(b), 3 marks

The time worked, T , in hours per week by employees of a large company is normally distributed with a mean of 42 and standard deviation 10.7.

A group of four employees is selected at random. Each employee is asked in turn whether they work more than 40 hours per week. Find the probability that the fourth employee is the only one in the group who works more than 40 hours per week.

The branch probability comes from the normal distribution — that is the GDC's work and it belongs to D5. What belongs here is the tree it sits in: three "no" and then one "yes", in that order and no other.

```
over = nsimplify(0.5741362)          # P(T >
40), from the normal distribution
four = {(i, j, m, n): (over if i else 1 -
over)*(over if j else 1 - over)
          *(over if m else 1 -
over)*(over if n else 1 - over)
          for i in (0, 1) for j in (0, 1) for m
in (0, 1) for n in (0, 1)}

q4_3 = ...

verify_probability('4.3', q4_3, four, lambda
o: o == (0, 0, 0, 1))
```

4.4 — May 2024 TZ2 Paper 1 Q4(a), 2 marks

A species of bird can nest in two seasons: Spring and Summer. The probability of nesting in Spring is k ; the probability of nesting in Summer is $\frac{k}{2}$.

Complete the tree diagram to show the probabilities of not nesting in each

season. Write your answers in terms of k .

Three boxes and two of them are the same. Write the Summer one.

```
q4_4 = ...          # the four path weights:
nest-nest, nest-not, not-nest, not-not

verify_constants('4.4', q4_4,
symbols('v1:5'), [
    ('the four paths together are certain',
Eq(sum(symbols('v1:5')), 1)),
    ('the bird nests in Spring with
probability k',
Eq(symbols('v1') + symbols('v2'), k_)),
    ('and in Summer with probability k/2
after nesting in Spring',
Eq(symbols('v1'), k_*k_/2)),
    ('with the same k/2 in Summer after not
nesting in Spring',
Eq(symbols('v3'), (1 - k_)*k_/2)),
])
```

5. Repeated trials and the first success

The same trial, over and over, independently. Two questions get asked: **how many successes** (that is D3) and **when the first one comes** (that is here).

$$P(\text{first success on the } n\text{th trial}) \\ = p(1 - p)^{n-1}$$

$n - 1$ failures, then a success. The exponent is $n - 1$ and not n , and that off-by-one is the whole difficulty of the section.

5.1 — May 2024 TZ1 Paper 2 Q6(c), 2 marks

In Happyland, the weather on any given day is independent of the weather on any other day. On any day in May, the probability of rain is 0.2. May has 31 days.

Find the probability that the first day that it rains in May is on the 10th day.

```
may = {n: Rational(2, 10)*Rational(8, 10)**(n
- 1) for n in range(1, 32)}
may['no rain at all'] = Rational(8, 10)**31

q5_1 = ...

verify_probability('5.1', q5_1, may, lambda
o: o == 10)
```

5.2 — May 2024 TZ1 Paper 3 Q1(a)(i), 2 marks

In a new computer game, each time a player performs an action, there is a random chance that the action will be *boosted*. In the first model, the probability that an action will be boosted is constant.

Suppose the probability that an action will be boosted is 0.1. Find the probability that the first boost occurs on the third action.

```
boost = {n: Rational(1, 10)*Rational(9, 10)**
(n - 1) for n in range(1, 400)}
boost['never'] = Rational(9, 10)**399

q5_2 = ...

verify_probability('5.2', q5_2, boost, lambda
o: o == 3)
```

5.3 — May 2024 TZ1 Paper 3 Q1(a)(ii), 3 marks

Find the probability that at least one boost occurs in the first six actions.

```
q5_3 = ...
```

```
verify_probability('5.3', q5_3, boost, lambda  
o: o != 'never' and o <= 6)
```

5.4 — May 2024 TZ1 Paper 3 Q1(e), 2 marks

In the designer's second model, the initial probability that an action is boosted is 0.2, and each time an action occurs that is not boosted, the probability that the next action is boosted increases by 0.2. After an action has been boosted, the probability resets to 0.2 for the next action.

Show that the probability that the first boost occurs on the third action is 0.288.

The printed answer is the work: writing the space down is the question, and the check tells you at once whether your space is the right one — it will not let the weights pass unless they come to one.

```

rising = [Rational(2, 10), Rational(4, 10),
          Rational(6, 10),
          Rational(8, 10), Rational(10, 10)]
second = {n: rising[n - 1]*Mul(*[1 - q for q
in rising[:n - 1]])
          for n in range(1, 6)}

q5_4 = ...

verify_probability('5.4', q5_4, second,
lambda o: o == 3)

```

5.5 — May 2024 TZ1 Paper 3 Q1(f), 1 mark · no cell

Let Y be the number of actions until the first boost occurs. Explain why $Y \leq 5$.

Nothing to hand over. Do it on paper and read the solution.

5.6 — May 2024 TZ1 Paper 3 Q1(g)(i), 2 marks

The following table shows the probability distribution of Y .

y	1	2	3	4	5
$P(Y = y)$	0.2	m	0.288	n	0.0384

Find the value of m and the value of n .

```

q5_6 = ...          # [m, n]

verify_constants('5.6', q5_6, symbols('m n'),
[
    ('the second action is boosted only if
the first was not',
    Eq(symbols('m'), Rational(8,
10)*Rational(4, 10))),
    ('and the fourth only if none of the
first three was',
    Eq(symbols('n'), Rational(8,
10)*Rational(6, 10)*Rational(4, 10)
    *Rational(8, 10))),
])

```

5.7 — May 2022 TZ2 Paper 1 Q10(c), 5 marks

A biased four-sided die with faces 1, 2, 3, 4 has $P(X = x)$ equal to 0.4, 0.3, 0.2, 0.1 respectively. Nicky plays a game with this die: she is allowed a maximum of five rolls, her score is the sum of the results, and she wins if her score is at least ten.

After three rolls of the die, Nicky has a score of four. Assuming that rolls of the die are independent, find the probability that Nicky wins the game.

Two rolls left and six more needed. The markscheme awards M0 to anyone who

also considers a score of 5 — the sum of the last two rolls has to reach 6, and 5 is not enough.

```
face = {1: Rational(4, 10), 2: Rational(3, 10),  
        3: Rational(2, 10), 4: Rational(1, 10)}  
last_two = {(u, v): face[u]*face[v] for u in  
            face for v in face}  
  
q5_7 = ...  
  
verify_probability('5.7', q5_7, last_two,  
                  lambda o: o[0] + o[1] >= 6)
```

6. The law of total probability

$$P(E) = \sum_i P(C_i) P(E \mid C_i)$$

The population splits into parts — two machines, three clerks, two kinds of apple — and you want a probability across the whole of it. Each part contributes its share times what happens inside it. The parts must cover everything and must not overlap; check that before anything else.

This is the section where the calculator is finally doing real work: the conditional probabilities inside the parts are normal or binomial, and they are not available on paper.

**6.1 — May 2022 TZ1 Paper 2 Q11(c)
(i), 4 marks**

A bakery makes chocolate muffins, whose weights C are normally distributed with mean 62 g and standard deviation 2.9 g, and banana muffins, whose weights B are normally distributed with mean 68 g and standard deviation 3.4 g. Each day 60% of the muffins made are chocolate.

On a particular day, a muffin is randomly selected from all those made at the bakery. Find the probability that the randomly selected muffin weighs less than 61 g.

```

light = {'chocolate': nsimplify(0.3651120),
'banana': nsimplify(0.0197555)}
muffin = {(kind, w): (Rational(6, 10) if kind
== 'chocolate' else Rational(4, 10))
            * (light[kind] if w ==
'<61' else 1 - light[kind])
            for kind in light for w in ('<61',
'>61')}}

q6_1 = ...

verify_probability('6.1', q6_1, muffin,
lambda o: o[1] == '<61')

```

6.2 — May 2025 TZ3 Paper 2 Q11(c) (i), 3 marks

Amanda, Bryce and Carmen enter data from surveys into a database. Surveys entered by Amanda, Bryce and Carmen are inaccurate 8%, 6% and 11% of the time respectively. From the surveys assigned to the three of them, Amanda enters 55%, Bryce 25% and Carmen 20%.

Find the probability that a randomly selected survey was entered inaccurately.

```

clerk = {'Amanda': (Rational(55, 100),
Rational(8, 100)),
        'Bryce': (Rational(25, 100),
Rational(6, 100)),
        'Carmen': (Rational(20, 100),
Rational(11, 100))}
survey = {(who, ok): share*(bad if ok ==
'wrong' else 1 - bad)
          for who, (share, bad) in
clerk.items() for ok in ('wrong', 'right')}

q6_2 = ...

verify_probability('6.2', q6_2, survey,
lambda o: o[1] == 'wrong')

```

6.3 — November 2025 TZ1 Paper 2 Q7(a), 4 marks

Two machines are used in a factory to manufacture semiconductors. Machine A manufactures defective semiconductors 10% of the time, while machine B manufactures defective semiconductors 5% of the time. A randomly selected machine manufactures ten semiconductors. Both machines are equally likely to be selected.

Find the probability that exactly two semiconductors are defective.

```
chip = {(name, d): Rational(1,
2)*binomial(10, d)*q**d*(1 - q)**(10 - d)
        for name, q in (('A', Rational(1,
10)), ('B', Rational(1, 20)))
        for d in range(11)}

q6_3 = ...

verify_probability('6.3', q6_3, chip, lambda
o: o[1] == 2)
```

7. Bayes: back up the tree

$$P(C \mid E) = \frac{P(C) P(E \mid C)}{P(E)}$$

The tree runs from cause to effect; the question runs the other way. The numerator is one path — the share of the cause times what it does — and the denominator is §6, the whole of $P(E)$. That is the entire method: the hard part was §6 and you have already done it.

The markscheme wants to see the conditional stated *in context*: «not just $P(A \mid B)$ ». Writing the formula with letters earns nothing.

7.1 — May 2022 TZ1 Paper 2 Q11(c) (ii), 3 marks

Given that a randomly selected muffin weighs less than 61 g, find the probability that it is chocolate.

```
q7_1 = ...
```

```
verify_probability('7.1', q7_1, muffin,  
                  lambda o: o[0] ==  
'chocolate', given=lambda o: o[1] == '<61')
```

7.2 — May 2024 TZ2 Paper 1 Q6(b), 4 marks

A farmer grows cooking apples and eating apples. The weights are normal: eating apples $\mu = 100$, $\sigma = 20$; cooking apples $\mu = 140$, $\sigma = 40$. For each type, 95% of the weights are within two standard deviations of the mean. 80% of the apples the farmer grows are eating apples. Both types are mixed together, and the machine separates out those weighing more than 140 g into a container.

An apple is randomly selected from this container. Find the probability that it is

an eating apple. Give your answer in the form $\frac{c}{d}$, where $c, d \in \mathbb{Z}^+$.

140 g is two standard deviations above the eating-apple mean and exactly the cooking-apple mean, so both branch probabilities are readable off the 95% rule without a calculator: 2.5% and 50%.

```
apple = {('eating', 'heavy'): Rational(8,
10)*Rational(25, 1000),
        ('eating', 'light'): Rational(8,
10)*(1 - Rational(25, 1000)),
        ('cooking', 'heavy'): Rational(2,
10)*Rational(1, 2),
        ('cooking', 'light'): Rational(2,
10)*Rational(1, 2)}
```

```
q7_2 = ...
```

```
verify_probability('7.2', q7_2, apple,
                  lambda o: o[0] ==
'eating', given=lambda o: o[1] == 'heavy')
```

7.3 — May 2025 TZ3 Paper 2 Q11(c) (ii), 3 marks

Find the probability that a randomly selected survey was entered by Amanda, given that the survey was entered inaccurately.

```
q7_3 = ...  
  
verify_probability('7.3', q7_3, survey,  
                  lambda o: o[0] ==  
'Amanda', given=lambda o: o[1] == 'wrong')
```

7.4 — November 2025 TZ1 Paper 2 Q7(b), 3 marks

Given that exactly two semiconductors are defective, find the probability that they were manufactured by machine A.

```
q7_4 = ...  
  
verify_probability('7.4', q7_4, chip,  
                  lambda o: o[0] == 'A',  
                  given=lambda o: o[1] == 2)
```

7.5 — November 2022 Paper 2 Q10(c) (i), 3 marks

The time worked, T , in hours per week is normally distributed with mean 42 and standard deviation 10.7. A large group of employees work more than 40 hours per week. An employee is selected at random from this large group. Find the probability that this employee works less than 55 hours per week.

The conditioning event is not one branch of a tree here — it is a piece of the real line. The machine is the same: the numerator is the overlap, the denominator is the condition.

```
hours = {'under 40': nsimplify(0.4258633),  
# from the normal distribution  
        '40 to 55': nsimplify(0.4619448),  
        'over 55': nsimplify(0.1121919)}  
  
q7_5 = ...  
  
verify_probability('7.5', q7_5, hours,  
                  lambda o: o != 'over 55',  
given=lambda o: o != 'under 40')
```

8. Without replacement

Nothing goes back in the box, so the denominator shrinks by one at every draw — and so does the numerator, if what you drew was what you wanted.

$$P(YY) = \frac{y}{r+y} \cdot \frac{y-1}{r+y-1}$$

Both shrink. Forgetting the numerator is the commoner slip of the two, because

the denominator is the one everybody has been told about.

8.1 — May 2023 TZ2 Paper 2 Q11(a), 4 marks

A game of chance involves drawing **two** balls at random out of a box without replacement. The box initially contains r red balls and y yellow balls. Let $P(YY)$ represent the probability of drawing two yellow balls from the box without replacement.

Consider a version of this game where it is known that $P(YY) = \frac{1}{3}$.

Show that $2y^2 - 2(r + 1)y + r - r^2 = 0$.

Write the equation your own algebra produced, before it was tidied into the printed form. The check accepts any equation that says the same thing up to a numerical factor and a rearrangement, which is what the markscheme accepts too — so this tells you whether your algebra landed, not whether you copied the line above.

```
q8_1 = ...          # your equation, in r_
and y_

verify_equation('8.1', q8_1,
                Eq(2*y_**2 - 2*(r_ + 1)*y_ +
r_ - r_**2, 0), var=y_)
```

8.2 — May 2023 TZ2 Paper 2 Q11(d), 3 marks

Now consider a similar game that involves drawing **three** balls out of a box without replacement. The box initially contains 10 red balls and y yellow balls.

Find an expression for $P(YYY)$ in terms of y .

```
q8_2 = ...          # in terms of y_

# the model, stage by stage: at draw i there
# are y - i yellow balls left
# out of y + 10 - i, and the answer is the
# product of the three
stages = [(y_ - i)/(y_ + 10 - i) for i in
range(3)]

verify_identity('8.2', q8_2, Mul(*stages),
var=y_, samples=(4, 7, 11, 20))
```

8.3 — May 2023 TZ2 Paper 2 Q11(e), 5 marks

A yellow ball is added so that the box now contains 10 red balls and $(y + 1)$ yellow balls. The probability of drawing three yellow balls from the box without replacement is now twice the probability expressed in part (d).

Find the initial number of yellow balls in the box.

```
q8_3 = ...          # the number of yellow
balls the box started with

verify_constants('8.3', [q8_3], [y_], [
    ('adding one yellow ball doubles the
probability of drawing three',
    Eq((y_ + 1)*y_*(y_ - 1)/((y_ + 11)*(y_ +
10)*(y_ + 9))),
    2*y_*(y_ - 1)*(y_ - 2)/((y_ + 10)*(y_
+ 9)*(y_ + 8))))),
1)
```

9. Counting an equally likely space

When the outcomes are equally likely and countable, probability is a fraction:

how many fit over how many there are. The work is entirely in the counting, and the counting is done by cases — never by listing.

Two things go wrong. The space is counted as *sets* when it is really *ordered tuples*: rolling 1, 2, 3 in that order is a different outcome from 3, 2, 1, and both are among the 216. And a case gets missed, which no amount of care with the arithmetic will fix.

9.1 — May 2024 TZ2 Paper 3 Q2(c), 6 marks

Consider quadratic functions $f(x) = ax^2 + bx + c$, whose coefficients a , b and c are randomly generated in turn by rolling an unbiased six-sided die three times and reading off the value shown on the uppermost face.

By considering the discriminant, or otherwise, show that the probability of the graph of such a randomly generated

quadratic function having only one x -intercept is $\frac{5}{216}$.

```
rolls = {(u, v, w): Rational(1, 216)
          for u in range(1, 7) for v in
          range(1, 7) for w in range(1, 7)}

q9_1 = ...

verify_probability('9.1', q9_1, rolls,
                  lambda t: t[1]**2 -
                  4*t[0]*t[2] == 0)
```

9.2 — May 2024 TZ2 Paper 3 Q2(f), 6 marks

Now consider randomly generated quadratic functions whose corresponding graphs have two **distinct** x -intercepts. Let p be the probability of the graph of such a randomly generated quadratic function having two distinct x -intercepts.

Using the approach started in part (e) — where $ac = 1$ gives four such functions and $ac = 2$ gives eight — or otherwise, find the value of p .

```
q9_2 = ...

verify_probability('9.2', q9_2, rolls,
                  lambda t: t[1]**2 -
4*t[0]*t[2] > 0)
```

9.3 — May 2022 TZ2 Paper 1 Q10(b), 2 marks

A biased four-sided die with faces labelled 1, 2, 3 and 4 is rolled. The probability distribution for X is p , 0.3, q , 0.1, and it is known that $E(X) = 2$; part (a) gives $p = 0.4$ and $q = 0.2$.

Find $P(X > 2)$.

```
biased = {1: Rational(4, 10), 2: Rational(3,
10),
          3: Rational(2, 10), 4: Rational(1,
10)}

q9_3 = ...

verify_probability('9.3', q9_3, biased,
                  lambda o: o > 2)
```

9.4 — May 2022 TZ2 Paper 1 Q10(d), 2 marks

David has two pairs of unbiased four-sided dice, a yellow pair and a red pair.

Both yellow dice have faces labelled 1, 2, 3 and 4; let S be the sum obtained by rolling them, so that $P(S = s)$ is $\frac{1}{16}, \frac{2}{16}, \frac{3}{16}, \frac{4}{16}, \frac{3}{16}, \frac{2}{16}, \frac{1}{16}$ for $s = 2, \dots, 8$.

The first red die has faces labelled 1, 2, 2 and 3. The second red die has faces labelled 1, a , a and b , where $a < b$ and $a, b \in \mathbb{Z}^+$. The probability distribution for the sum obtained by rolling the red pair is the same as the distribution for the yellow pair.

Determine the value of b .

```
q9_4 = ...
```

```
verify_constants('9.4', [q9_4], [b_], [
    ('the largest sum the red pair can make
    is the largest the yellow '
    'pair can make', Eq(3 + b_, 8)),
])
```

9.5 — May 2022 TZ2 Paper 1 Q10(e), 2 marks

Find the value of a , providing evidence for your answer.

One equation says everything here. The distribution of a sum is the product of the

two dice's generating polynomials — a face f contributes z^f — so two pairs of dice give the same sums exactly when the two products agree as polynomials. Hand over both faces and the identity settles them together.

```
q9_5 = ...                # [a, b]

verify_constants('9.5', q9_5, [a_, b_], [
    ('the red pair and the yellow pair have
the same generating polynomial',
    Eq(expand((z_ + 2*z_**2 + z_**3)*(z_ +
2*z_**a_ + z_**b_)),
        expand((z_ + z_**2 + z_**3 +
z_**4)**2))),
    ])
```

10. A letter inside the probability

The space itself has an unknown in it — a proportion k , a probability p , a percentage x — and a probability computed from it is given. You do not substitute; you solve.

And then you throw a root away. Not because it is extraneous in the usual

algebraic sense, but because it is not a probability: it is bigger than one or less than zero. **Say so out loud** — the markscheme gives a whole mark for the sentence, and gives nothing for the right answer arrived at silently.

10.1 — May 2024 TZ2 Paper 1 Q4(b), 4 marks

A species of bird nests in Spring with probability k and in Summer with probability $\frac{k}{2}$, independently. It is known that the probability of not nesting in Spring and not nesting in Summer is $\frac{5}{9}$.

(i) Show that $9k^2 - 27k + 8 = 0$. (ii) Both $k = \frac{1}{3}$ and $k = \frac{8}{3}$ satisfy $9k^2 - 27k + 8 = 0$. State why $k = \frac{1}{3}$ is the only valid solution.

Hand over the value of k ; the check puts it back into the sentence the question actually states, which is (i) done properly. What (ii) wants is one line of prose, and the solution gives it.

```

q10_1 = ...          # [k]

verify_constants('10.1', q10_1, [k_], [
    ('not nesting in Spring and not nesting
in Summer has probability 5/9',
    Eq((1 - k_)*(1 - k_/2), Rational(5,
9))),
], domain=Interval(0, 1))

```

10.2 — May 2023 TZ1 Paper 2 Q7(b), 4 marks

A test returns a correct positive result 95% of the time for parrots with the gene G , and a false positive result 2% of the time for parrots without it. The proportion of the population with the gene is p .

A random sample of the population was tested. It was found that 150 tests returned a positive result. Out of the 150 parrots with a positive test result, 18 did not actually have the gene. Find an estimate for p .

The 18 out of 150 is a conditional probability, and the conditioning is on the test being positive — not on the parrot. That is the whole question.

```

q10_2 = ...          # [p], to three
significant figures

verify_constants('10.2', q10_2, [p_], [
    ('18 of the 150 positive tests were on
    parrots without the gene',
    Eq(Rational(2, 100)*(1 - p_)
        / (Rational(95, 100)*p_ + Rational(2,
100)*(1 - p_)),
        Rational(18, 150))),
], tol=5e-6)

```

10.3 — May 2025 TZ3 Paper 2 ***Q11(d), 4 marks***

The following year, the accuracy of Amanda's and Bryce's work remained the same (8% and 6% inaccurate), as did the percentage of surveys entered by each of the three employees (55%, 25%, 20%). However, Carmen's accuracy had improved and the probability that she entered a survey inaccurately was now $x\%$.

The probability that a randomly selected survey had been entered inaccurately was now the same as the probability that Carmen made an error when entering a survey.

Find the value of x .

```
q10_3 = ...          # [x], as a percentage

verify_constants('10.3', q10_3, [a_], [
    ('the overall inaccuracy equals Carmen`s
    own',
        Eq(Rational(55, 100)*Rational(8, 100) +
            Rational(25, 100)*Rational(6, 100)
            + Rational(20, 100)*a_/100, a_/100)),
    ])
```



Solutions

Numbered as above. Open one after you have worked the question.

1. The algebra of events

1.1 $P(A \cup B) = 0.65 + 0.75 - 0.6 = \boxed{0.8}$. Or read it off the diagram: the three cells $0.05 + 0.6 + 0.15$.

1.2 By De Morgan, $A' \cap B' = (A \cup B)'$, so $P = 1 - 0.8 = \boxed{0.2}$. The markscheme note is worth reading: *"0.2 must be stated as the candidate's answer, or labeled as $P(A' \cap B')$ in their Venn diagram. Just seeing an unlabeled 0.2 in the correct region earns M1A0."* A number in the right place is not an answer.

1.3 $P(S) + P(T) + P(S' \cap T') - P(S \cap T) = 1$, so $0.7 + 0.2 + 0.18 - P(S \cap T) = 1$ and $P(S \cap T) = \boxed{0.08}$. Equivalently $P(S \cup T) = 1 - 0.18 = 0.82$ and then the addition rule.

1.4 $P(T \cap S') = P(T) - P(T \cap S) = 0.2 - 0.08 = \boxed{0.12}$.

1.5 $A \cup B$ splits into $A \cap B'$ and B , which do not overlap: $\frac{5}{8} = \frac{7}{24} + P(B)$, so $P(B) = \frac{15-7}{24} = \boxed{\frac{1}{3}}$.

1.6 Independent means $P(A \cap B) = P(A)P(B) = 0.3 \times 0.8 = \boxed{0.24}$.

1.7 $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 1.1 - P(A \cap B)$, and a probability is at most 1, so $P(A \cap B) \geq 0.1$. The value 0.1 is attained when $A \cup B$ is everything. $\boxed{0.1}$

1.8 $P(A \cap B) \leq P(A) = 0.3$, attained when $A \subseteq B$ — which is possible here because $P(A) < P(B)$. $\boxed{0.3}$

The reasoning is the mark. The markscheme wants "*A is a subset of B (so $P(A \cap B) = P(A)$)*", or a clearly labelled diagram with A entirely inside B , and it says explicitly **"Do not award R0A1"** — the number alone scores nothing.

2. Conditional probability

$$2.1 \quad P(A \cap B) = P(A \mid B) P(B) = \frac{1}{4} \cdot \frac{1}{3} = \boxed{\frac{1}{12}}.$$

Note which way the multiplication goes: the conditional multiplies the probability of *its own condition*.

$$2.2 \quad P(B') = \frac{2}{3} \quad \text{and} \quad P(A \cap B') = P(A) - P(A \cap B) = \frac{1}{2} - \frac{1}{12} = \frac{5}{12}. \quad \text{Then} \quad P(A \mid B') = \frac{5/12}{2/3} = \boxed{\frac{5}{8}}.$$

Not $1 - P(A \mid B) = \frac{3}{4}$. The complement is taken of the *event*, and here the complement is taken of the *condition* — a different operation entirely, and the two agree only by accident.

$$2.3 \quad P(A) = 1 - \frac{3}{4} = \frac{1}{4}, \quad \text{and} \quad P(A \cap B) = P(B \mid A) P(A) = \frac{2}{3} \cdot \frac{1}{4} = \boxed{\frac{1}{6}}.$$

2.4 Two equations in $P(B)$ and $P(A \cap B)$:

$$\begin{aligned} 0.55 &= 0.4 + P(B) - P(A \cap B), & 0.25 \\ &= \frac{P(A \cap B)}{P(B)}. \end{aligned}$$

Substituting the second into the first:
 $0.55 = 0.4 + P(B) - 0.25P(B)$, so $0.75P(B) = 0.15$ and $P(B) = \boxed{0.2}$.

2.5 Because A and B are independent, so are A' and B' , and therefore $P(A' | B') = P(A')$. From part (a), $P(A \cap B) = 0.24$, so $P(A) = 0.16 + 0.24 = 0.4$ and $P(A') = \boxed{0.6}$.

The long way works too: $P(B') = 1 - 0.6 = 0.4$, $P(A' \cap B') = 1 - 0.16 - 0.24 - 0.36 = 0.24$, and $0.24/0.4 = 0.6$.

2.6 From 1.5, $P(B) = \frac{1}{3}$. Independence gives $P(A \cap B') = P(A)P(B')$, so $\frac{7}{24} = P(A) \cdot \frac{2}{3}$ and $P(A) = \frac{7}{16}$. Then $P(A' | B')$ — no: the question asks $P(A' | B)$, and independence again makes the condition irrelevant: $P(A' | B) = P(A') = 1 - \frac{7}{16} = \boxed{\frac{9}{16}}$.

2.7 $P(G \cap T) = P(T | G)P(G) = 0.25 \times 0.48 = \boxed{0.12}$.

Notice that this equals $P(T \cap S')$ from 1.4 — a coincidence of this paper, and a good reminder that equal numbers are not the same event.

3. Independence as an equation

3.1 From 2.3, $P(A) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{6}$. The addition rule gives $\frac{3}{4} = \frac{1}{4} + P(B) - \frac{1}{6}$, so $P(B) = \frac{2}{3}$. Then

$$P(A)P(B) = \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{6} = P(A \cap B),$$

so A and B are independent. $\boxed{[\frac{1}{6}, \frac{1}{6}]}$

The markscheme note: "*The R1 is dependent on all previous marks.*" The conclusion is only worth something standing on the numbers.

3.2 $P(G)P(T) = 0.48 \times 0.2 = 0.096$, while $P(G \cap T) = 0.12$ from 2.7. They differ, so G and T are **not** independent. $\boxed{[0.096, 0.12]}$

Or, equivalently, $P(T | G) = 0.25 \neq 0.2 = P(T)$: knowing the student is a girl changes the probability of theatre, and that is what dependence means.

3.3 $P(A) = \frac{1}{2}$; $P(R) = \frac{9}{14}$ from 4.2; $P(A \cap R) = \frac{1}{2} \cdot \frac{5}{7} = \frac{5}{14}$. Then

$$P(A)P(R) = \frac{1}{2} \cdot \frac{9}{14} = \frac{9}{28} \neq \frac{5}{14} = \frac{10}{28},$$

so **not** independent. $\boxed{[\frac{9}{28}, \frac{5}{14}]}$

The markscheme also accepts prose — *"different number of red balls in each box"* — but insists: *"Both conclusion and reasoning are required. Do not split the A2."*

3.4 Let $P(A \cap B) = x$. Then $P(A) = x + 0.16$ and $P(B) = x + 0.36$, and independence says

$$\begin{aligned}x &= (x + 0.16)(x + 0.36) \\&\implies x^2 - 0.48x + 0.0576 \\&= 0 \\&\implies (x - 0.24)^2 \\&= 0,\end{aligned}$$

so $x = \boxed{0.24}$. A repeated root — the two events are forced.

3.5 Write $P(B) = b$, so $P(A) = 3b$. Independence turns the addition rule into

$$\begin{aligned}
0.68 &= 3b + b - 3b^2 \\
&\implies 3b^2 - 4b + 0.68 \\
&= 0 \\
&\implies b \\
&= 0.2 \text{ or } b \\
&= \frac{17}{15}.
\end{aligned}$$

The second is $1.133\dots > 1$, and a probability cannot exceed one, so $P(B) = \boxed{0.2}$.

The markscheme awards A2 for 0.2 alone and only **A1** if both are offered as final answers. Rejecting the root is half the mark.

4. Trees

4.1 The four paths:

$$0.95p, \quad 0.05p, \quad 0.02(1 - p), \quad 0.98(1 - p).$$

The missing labels are $1 - p$ on the lower first branch and 0.95, 0.05 on the two branches out of G . The markscheme insists the first be written as $1 - p$:
"award A0 for G' branch labelled q unless

explicitly defined as $1 - p$ ". A letter is not a probability until it is defined.

$$\mathbf{4.2} \quad P(R) = P(R \cap B_1) + P(R \cap B_2) = \frac{1}{2} \cdot \frac{5}{7} + \frac{1}{2} \cdot \frac{4}{7} = \frac{5}{14} + \frac{4}{14} = \boxed{\frac{9}{14}}.$$

4.3 $P(T > 40) = 0.574136 \dots$ from the normal distribution. "Only the fourth" is one specific order — no, three orders are not wanted here: the question names *the fourth* employee, so it is one path:

$$(1 - 0.574136)^3 \times 0.574136 = \boxed{0.0443}.$$

Compare with 5.7, where the order is free and the paths must be counted.

4.4 Not nesting in Spring is $1 - k$; not nesting in Summer is $1 - \frac{k}{2}$, and it is the same on both branches because the seasons are independent. The four paths:

$$\frac{k^2}{2},$$

$$k\left(1 - \frac{k}{2}\right),$$

$$\frac{(1-k)k}{2},$$

$$(1-k)\left(1 - \frac{k}{2}\right).$$

5. Repeated trials and the first success

5.1 Nine dry days and then rain: $0.8^9 \times 0.2 = \boxed{0.0268}$.

Not $0.8^{10} \times 0.2$ (that is the first rain on the eleventh day) and not 0.2 (that is rain on the tenth day regardless of what came before).

5.2 $0.9^2 \times 0.1 = \boxed{0.081}$.

5.3 Through the complement: $1 - 0.9^6 = 1 - 0.531441 = \boxed{0.469}$.

Adding six terms gives the same number and takes six times as long; and the sixth term is the one people drop.

5.4 In the second model the probability of a boost is 0.2, then 0.4, then 0.6, ... until it happens. The first boost on the third action needs two misses and then a hit:

$$0.8 \times 0.6 \times 0.6 = \boxed{0.288}.$$

The third factor is 0.6 and not 0.4: after two misses the probability has risen twice.

5.5 After four consecutive misses the probability of a boost has risen to $0.2 + 4(0.2) = 1$, so the fifth action is boosted with certainty. Hence $Y \leq 5$. (One mark, one sentence.)

5.6 $m = P(Y = 2) = 0.8 \times 0.4 = \boxed{0.32}$ and
 $n = P(Y = 4) = 0.8 \times 0.6 \times 0.4 \times 0.8 = \boxed{0.1536}$

Check: $0.2 + 0.32 + 0.288 + 0.1536 + 0.0384 = 1$. That is worth doing — it catches a dropped factor instantly.

5.7 Nicky has 4 after three rolls and two rolls left, so she needs at least 6 from two rolls. The markscheme is explicit:

"Award M0 if candidate also considers scores other than 6, 7, or 8 (such as 5)."

$$P(6) = 2(0.3)(0.1) + (0.2)^2 = 0.1,$$

$$P(7) = 2(0.2)(0.1) = 0.04,$$

$$P(8) = (0.1)^2 = 0.01$$

$$P(\text{wins}) = 0.1 + 0.04 + 0.01 = \boxed{0.15}$$

The factors of 2 are the orders: (2, 4) and (4, 2) are different rolls.

6. The law of total probability

6.1 $P(C < 61) = 0.365112$ and $P(B < 61) = 0.0197555$ from the normal distributions. Weighting by the shares:

$$0.6 \times 0.365112 + 0.4 \times 0.0197555 = \boxed{0.227}.$$

6.2 $0.55(0.08) + 0.25(0.06) + 0.20(0.11) = 0.044 + 0.015 + 0.022 = \boxed{0.081}.$

The shares add to one — check that before anything else. Adding the three error rates instead (0.25) is the standard wrong answer, and it is wrong by a factor of three.

6.3 Each machine gives a binomial: $B(10, 0.1)$ and $B(10, 0.05)$.

$$0.5 \times 0.193710 + 0.5 \times 0.0746347 = \boxed{0.134}.$$

7. Bayes: back up the tree

7.1

$$P(\text{choc} \mid <61) = \frac{0.6 \times 0.365112}{0.226969} = \boxed{0.965}.$$

Almost all light muffins are chocolate, which makes sense: the banana ones are heavier and there are fewer of them.

7.2 $P(\text{weight} > 140 \mid \text{eating}) = 0.025$, because 140 g is two standard deviations above 100 and 95% lie within two, leaving 2.5% above. $P(\text{weight} > 140 \mid \text{cooking}) = 0.5$, because 140 is the cooking mean. So

$$\frac{0.8 \times 0.025}{0.8 \times 0.025 + 0.2 \times 0.5} = \frac{0.02}{0.12} = \boxed{\frac{1}{6}}.$$

Four fifths of the apples are eating apples and still only one apple in six in the container is one — that is what conditioning does.

7.3

$$\frac{0.55 \times 0.08}{0.081} = \frac{0.044}{0.081} = \boxed{0.543}.$$

Amanda enters 55% of the surveys and makes 54% of the errors: her rate is close to the average, so conditioning barely moves the share.

7.4

$$\frac{0.5 \times 0.193710}{0.134172} = \boxed{0.722}.$$

The markscheme accepts 0.723 or 0.724 from three-significant-figure values carried through — but only because it says so; do not count on that.

7.5

$$\begin{aligned} P(T < 55 \mid T > 40) &= \frac{P(40 < T < 55)}{P(T > 40)} \\ &= \frac{0.461944}{0.574136} \\ &= \boxed{0.805}. \end{aligned}$$

The markscheme note here is a small mercy worth knowing: *"Do not penalize for inclusion or non-inclusion of endpoints for probabilities using a*

normal distribution." For a continuous variable the endpoint has probability zero, so $\leq$ and $<$ genuinely agree.

8. Without replacement

8.1

$$P(YY) = \frac{y}{r+y} \cdot \frac{y-1}{r+y-1} = \frac{1}{3}$$

$$3y(y-1) = (r+y)(r+y-1)$$

$$3y^2 - 3y = r^2 + 2ry + y^2 - r - y$$

$$2y^2 - 2ry - 2y + r - r^2 = 0$$

$$\implies \boxed{2y^2 - 2(r+1)y + r - r^2 = 0}$$

8.2 Three yellows out of y yellow and 10 red:

$$P(YYY) = \boxed{\frac{y(y-1)(y-2)}{(y+10)(y+9)(y+8)}}$$

8.3 With one more yellow ball the box holds $y+1$ yellow and 10 red:

$$\frac{(y+1)y(y-1)}{(y+11)(y+10)(y+9)} \\ = 2 \cdot \frac{y(y-1)(y-2)}{(y+10)(y+9)(y+8)}$$

Cancel $y(y-1)$ from both sides — legitimate, since $y \geq 3$ for the original probability to be non-zero — and then $(y+10)(y+9)$. The markscheme awards a method mark for reaching exactly this line:

$$\frac{y+1}{y+11} = \frac{2(y-2)}{y+8} \\ (y+1)(y+8) = 2(y-2)(y+11) \\ \implies y^2 + 9y + 8 \\ = 2y^2 + 18y - 44 \\ y^2 + 9y - 52 = 0 \\ \implies (y-4)(y+13) \\ = 0 \\ \implies \boxed{y=4}$$

The negative root goes for the usual reason — a box cannot hold -13 balls. Worth one line of checking: with four yellow, $P(YYY) = \frac{4 \cdot 3 \cdot 2}{14 \cdot 13 \cdot 12} = \frac{1}{91}$; with five, $\frac{5 \cdot 4 \cdot 3}{15 \cdot 14 \cdot 13} = \frac{2}{91}$. Twice, as promised.

9. Counting an equally likely space

9.1 One x -intercept means the discriminant is zero: $b^2 = 4ac$. With $b \in \{1, \dots, 6\}$, $b^2 \in \{1, 4, 9, 16, 25, 36\}$ and $4ac$ is a multiple of 4, so b must be even: $b = 2$ gives $ac = 1$, $b = 4$ gives $ac = 4$, $b = 6$ gives $ac = 9$.

- $ac = 1$: $(1, 1)$ — one way.
- $ac = 4$: $(1, 4), (2, 2), (4, 1)$ — three ways.
- $ac = 9$: $(3, 3)$ — one way. ($(1, 9)$ and $(9, 1)$ need a 9.)

Five triples out of 216: $\boxed{\frac{5}{216}}$.

9.2 Two distinct intercepts means $b^2 > 4ac$, that is $b > 2\sqrt{ac}$. Since $b \leq 6$, only $ac \leq 8$ can contribute. Count by the value of ac : how many ordered pairs (a, c) give it, and how many b clear the bar.

ac	$2\sqrt{ac}$	(a, c) pairs	b that work	product
1	2	$(1, 1) - 1$	$3, 4, 5, 6 - 4$	4
2	2.83	$(1, 2), (2, 1) - 2$	$3, 4, 5, 6 - 4$	8
3	3.46	$(1, 3), (3, 1) - 2$	$4, 5, 6 - 3$	6
4	4	$(1, 4), (2, 2), (4, 1) - 3$	$5, 6 - 2$	6
5	4.47	$(1, 5), (5, 1) - 2$	$5, 6 - 2$	4
6	4.90	$(1, 6), (2, 3), (3, 2), (6, 1) - 4$	$5, 6 - 2$	8
8	5.66	$(2, 4), (4, 2) - 2$	$6 - 1$	2

$$4 + 8 + 6 + 6 + 4 + 8 + 2 = 38, \text{ so } p = \frac{38}{216} =$$

$$\boxed{\frac{19}{108}} \approx 0.176.$$

Two rows are easy to get wrong. At $ac = 4$ the bar is exactly 4, and $b = 4$ gives a repeated root, not two distinct ones — that case belongs to 9.1. At $ac = 6$ there are four pairs, not two: 6 factorises as 1×6 and 2×3 , and both orders count. This is why the check counts the 216 triples itself instead of trusting a table.

9.3 $P(X > 2) = q + 0.1 = 0.2 + 0.1 = \boxed{0.3}$.

9.4 The largest sum the yellow pair can make is 8. The red pair's largest is $3 + b$, so $3 + b = 8$ and $b = \boxed{5}$.

9.5 The distribution of a sum is the product of generating polynomials: a die with faces f_i contributes $\sum z^{f_i}$.

$$\begin{aligned}\text{yellow: } (z + z^2 + z^3 + z^4)^2 &= z^2(1 + z + z^2 + z^3)^2 \\ &= z^2(1 + z)^2(1 + z^2)^2\end{aligned}$$

$$\begin{aligned}\text{red: } (z + 2z^2 + z^3)(z + 2z^a + z^b) \\ = z(1 + z)^2(z + 2z^a + z^b)\end{aligned}$$

Setting them equal: $z + 2z^a + z^b = z(1 + z^2)^2 = z + 2z^3 + z^5$, so $a = \boxed{3}$ and $b = 5$, confirming 9.4.

Without the polynomial the evidence is a table of the sixteen sums with $a = 3$: they come out 1, 2, 3, 4, 3, 2, 1 over $s = 2, \dots, 8$, matching the yellow row exactly. Either is accepted; the polynomial is shorter and proves there is no other answer.

10. A letter inside the probability

10.1 (i) Not nesting in either season:

$$\begin{aligned}(1 - k) \left(1 - \frac{k}{2} \right) &= \frac{5}{9} \\ \implies 1 & \\ &- k \\ &- \frac{k}{2} \\ &+ \frac{k^2}{2} \\ &= \frac{5}{9}\end{aligned}$$

$$\implies 18 - 18k - 9k + 9k^2 = 10$$

$$\implies \boxed{9k^2 - 27k + 8 = 0}$$

(ii) $k = \frac{8}{3} > 1$, and k is a probability, so it cannot exceed one. Hence $k = \boxed{\frac{1}{3}}$.

That sentence is the R1. The markscheme accepts "any valid reasoning indicating that any probability cannot be greater than 1 and/or probability cannot be less than 0" — and nothing else.

10.2 Of the parrots that test positive, the fraction without the gene is

$$\begin{aligned} P(G' \mid +) &= \frac{0.02(1 - p)}{0.95p + 0.02(1 - p)} \\ &= \frac{18}{150} \\ &= 0.12. \end{aligned}$$

Solving: $0.02 - 0.02p = 0.12(0.95p + 0.02)$, so
 $0.02 - 0.02p = 0.1116p + 0.0024$, giving
 $0.1316p = 0.0176$ and $p = \boxed{0.134}$.

The other route: $0.95pS = 132$ and $0.02(1 - p)S = 18$ for the sample size S ; dividing removes S and gives the same equation.
 $S \approx 1039$.

10.3 With Carmen's rate $x\%$,

$$\begin{aligned} P(I) &= 0.55(0.08) \\ &\quad + 0.25(0.06) \\ &\quad + 0.20\left(\frac{x}{100}\right) \\ &= \frac{x}{100} \end{aligned}$$

$$\begin{aligned}
0.059 + 0.002x &= 0.01x \\
&\implies 0.008x \\
&= 0.059 \\
&\implies x \\
&= \boxed{7.38}
\end{aligned}$$

Read the condition twice before writing it: it says the *overall* inaccuracy equals *Carmen's own*, which is why $x/100$ appears on both sides. It does not say her rate is the average of the three.